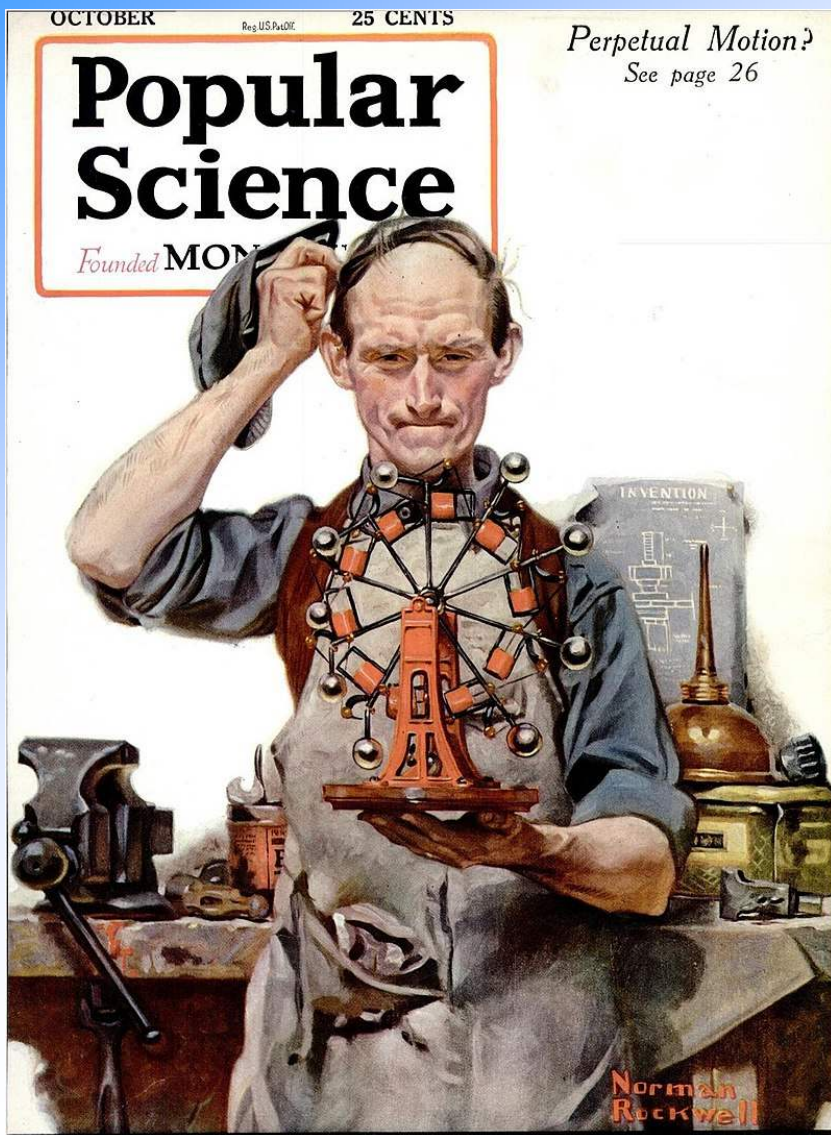


回顾:



第一类永动机是不可能制造的!

----热力学第一定律

$$Q = \Delta E + W$$

$$dQ = dW + dE \quad \text{微分形式}$$

对理想气体:

$$\left\{ \begin{array}{l} W = \int_{V_1}^{V_2} P dV \\ \Delta E = \frac{i}{2} \nu R \Delta T \\ Q = \nu \int_{T_1}^{T_2} C_m dT \end{array} \right.$$
$$C = \frac{dQ}{dT}$$
$$C_{V,m} = \frac{i}{2} R \quad C_{p,m} = C_{V,m} + R$$
$$\gamma = \frac{C_{p,m}}{C_{V,m}}$$

第3节 热力学第一定律对理想气体的应用

Applying the First Law of Thermodynamics to Ideal Gas

一、等容过程

特征 $dV=0$, $dW=0$

过程方程 $V=C_1$ 或 $\frac{p}{T}=C_2$

过程中吸热 $(dQ)_V = dE$

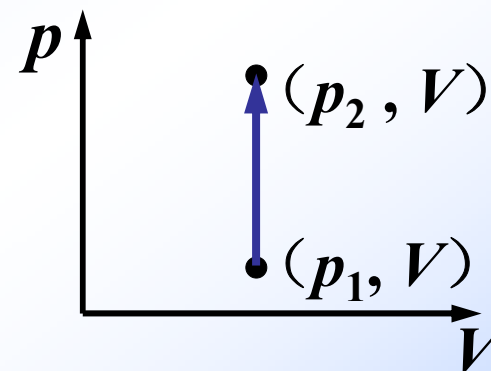
内能增量 $\Delta E = \nu C_{V,m}(T_2 - T_1) = \nu \frac{i}{2} R \Delta T$

则 $Q = \Delta E = \nu C_{V,m}(T_2 - T_1)$

或 $Q = \int dQ = \int_{T_1}^{T_2} \nu C_{V,m} dT = \nu C_{V,m}(T_2 - T_1)$

对外做功 $W = 0$

结论：等容过程系统吸收的热量全部用来增加内能。



二、等温过程

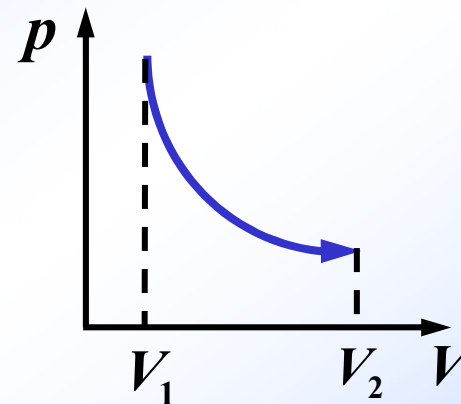
设 ν 摩尔理想气体经历等温过程

特征 $dT = 0$

过程方程 $T = C_1$ 或 $pV = C_2$

内能增量 $\Delta E = 0$

过程中吸热 $dQ = dW$



$$\begin{aligned}(Q)_T &= W = \int_{V_1}^{V_2} p dV \\&= \int_{V_1}^{V_2} \frac{1}{V} \nu R T dV = \nu R T \ln \frac{V_2}{V_1} \\&\because p_1 V_1 = p_2 V_2 \quad \therefore W = \nu R T \ln \frac{p_1}{p_2}\end{aligned}$$

结论：系统吸收的热量全部用来对外做功。

三、等压过程

设 ν 摩尔理想气体经历等压过程

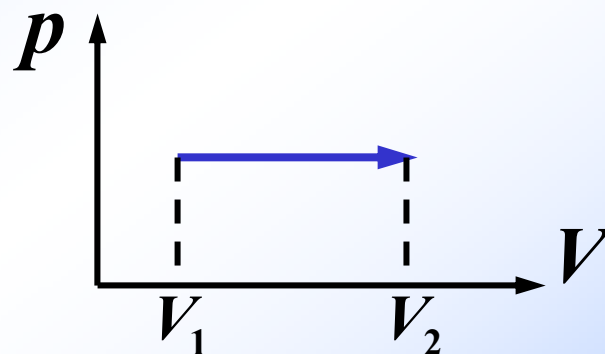
$$C_{p,m} = \frac{i+2}{2}R$$



特征 $dp = 0$

过程方程 $p = C_1$ 或 $\frac{V}{T} = C_2$

过程中吸热



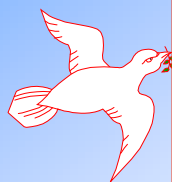
$$(Q)_p = \int_{T_1}^{T_2} \nu C_{p,m} dT = \nu \frac{i+2}{2} R (T_2 - T_1)$$

对外做功

$$W = \int_{V_1}^{V_2} p dV = p(V_2 - V_1) = \nu R (T_2 - T_1)$$

内能增量

$$\Delta E = (Q)_p - W = \nu \frac{i}{2} R (T_2 - T_1) = \nu C_{V,m} (T_2 - T_1)$$



$$Q_p = \nu C_{p,m}(T_2 - T_1)$$

$$W = \nu R(T_2 - T_1)$$

$$\Delta E = \nu C_{V,m}(T_2 - T_1)$$

结论

1° 等压过程中，系统吸收的热量一部分用来增加内能，一部分用来对外做功。

2° 在等容和等压两个等值过程中，均有

$$\Delta E = \nu C_{V,m}(T_2 - T_1)$$

ΔE 与过程无关，与过程是否为准静态过程也没有关系，它是理想气体内能增量的普遍式。

例10-1 将500J的热量传给标准状态下的2mol氢.

(1) V 不变, 热量变为什么? 氢的温度为多少?

(2) T 不变, 热量变为什么? 氢的 p 、 V 各为多少?

(3) p 不变, 热量变为什么? 氢的 T 、 V 各为多少?

解: (1) V 不变, $Q = \Delta E$, 热量转变为内能

$$\Delta E = Q_V = \frac{m}{M} C_{V,m} (T - T_0) = \nu \frac{5}{2} R (T - T_0)$$

$$\begin{aligned} T &= \frac{2Q_V}{5\nu R} + T_0 = \frac{2 \times 500}{5 \times 2 \times 8.31} + 273 \\ &= 285(\text{K}) \end{aligned}$$

例10-1 将500J的热量传给标准状态下的2mol氢.

(1) V 不变, 热量变为什么? 氢的温度为多少?

(2) T 不变, 热量变为什么? 氢的 p 、 V 各为多少?

(3) p 不变, 热量变为什么? 氢的 T 、 V 各为多少?

解: (2) T 不变, $Q = W$, 热量转变为功

$$Q = W = \frac{m}{M} RT \ln \frac{p_0}{p} \quad \Rightarrow \quad \frac{Q}{\nu RT} = \ln \frac{p_0}{p}$$

$$p = p_0 e^{-\frac{Q}{\nu RT}} = 1.013 \times 10^5 \times e^{-\frac{500}{2 \times 8.31 \times 273}} = 9.07 \times 10^4 (\text{Pa})$$

$$V = \frac{p_0 V_0}{p} = \frac{\nu RT}{p} = \frac{2 \times 8.31 \times 273}{9.07 \times 10^4} = 5.00 \times 10^{-2} (\text{m}^3)$$

例10-1 将500J的热量传给标准状态下的2mol氢.

(1) V 不变, 热量变为什么? 氢的温度为多少?

(2) T 不变, 热量变为什么? 氢的 p 、 V 各为多少?

(3) p 不变, 热量变为什么? 氢的 T 、 V 各为多少?

解:(3) p 不变, $Q = W + \Delta E$, 热量转变为功和内能

$$Q_p = \frac{m}{M} C_{p,m} (T - T_0) = 2 \times \frac{7}{2} R (T - T_0)$$

$$T = \frac{Q_p}{7R} + T_0 = \frac{500}{7 \times 8.31} + 273 \text{K} = 281.6 \text{(K)}$$

$$V_0 = \frac{\nu RT}{p_0} = \frac{2 \times 8.31 \times 273}{1.013 \times 10^5} = 44.8 \times 10^{-3} \text{(m}^3\text{)}$$

$$V = \frac{V_0 T}{T_0} = \frac{44.8 \times 10^{-3} \times 281.6}{273} \text{m}^3 = 0.046 \text{(m}^3\text{)}$$

例10-2 质量为 $2.8 \times 10^{-3} \text{kg}$ 、压强为 $1.013 \times 10^5 \text{Pa}$ 、温度为 27°C 的氮气, 先在体积不变的情况下使其压强增至 $3.039 \times 10^5 \text{Pa}$, 再经等温膨胀使压强降至 $1.013 \times 10^5 \text{Pa}$, 然后又在等压过程中将体积压缩一半. 试求氮气在全部过程中的内能变化, 所作的功以及吸收的热量, 并画出 p - V 图.

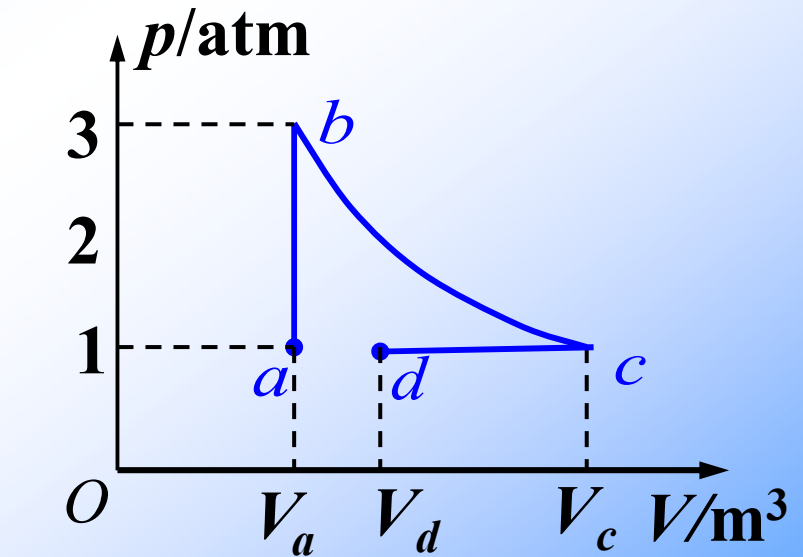
解: $T_a = 300\text{K}$ $p_a = 1.013 \times 10^5 \text{Pa}$

$$V_a = \frac{mRT_a}{Mp_a} = 2.46 \times 10^{-3} \text{m}^3$$

$$V_b = V_a, p_b = 3.039 \times 10^5 \text{Pa}$$

$$T_b = \frac{p_b}{p_a} T_a = 900\text{K}$$

$$V_c = \frac{p_b V_b}{p_c} = 7.38 \times 10^{-3} \text{m}^3$$



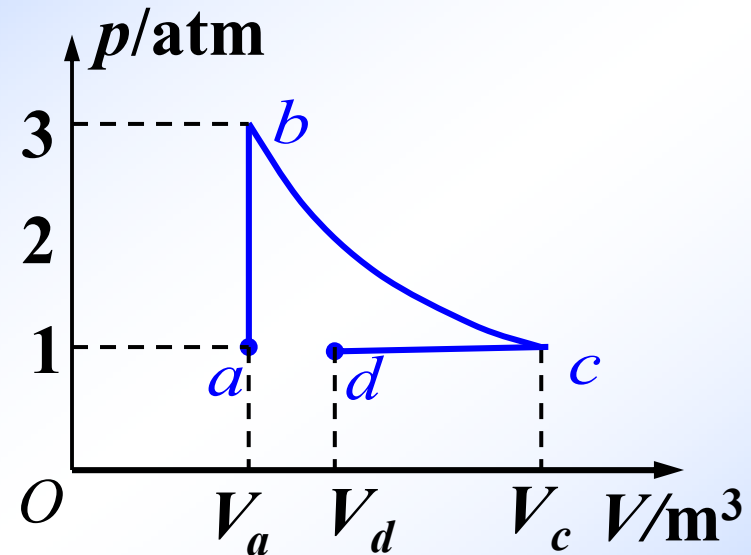
$$T_c = 900\text{K}$$

$$p_c = 1.013 \times 10^5 \text{Pa}$$

$$p_d = p_c = 1.013 \times 10^5 \text{ Pa}$$

$$V_d = \frac{V_c}{2} = 3.69 \times 10^{-3} \text{ m}^3$$

$$T_d = \frac{V_d}{V_c} T_c = 450 \text{ K}$$



等体过程: $W_1 = 0$

$$Q_1 = \Delta E_1 = \frac{m}{M} \frac{5}{2} R (T_b - T_a) = 1248 \text{ J}$$

等温过程: $\Delta E_2 = 0$ $Q_2 = W_2 = \frac{m}{M} R T_b \ln \frac{V_c}{V_b}$

$$= 823 \text{ J}$$

等压过程:

$$W_3 = p_c(V_d - V_c) = -374 \text{ J}$$

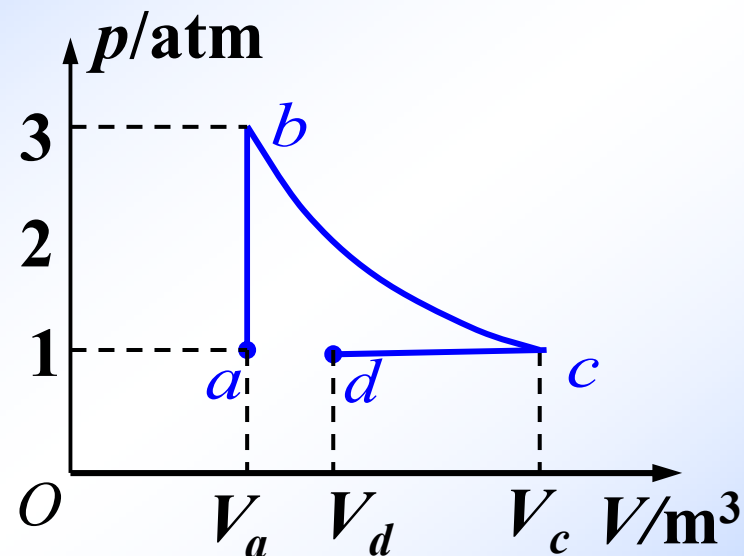
$$\Delta E_3 = \frac{m}{M} \frac{5}{2} R(T_d - T_c) = -936 \text{ J}$$

$$Q_3 = W_3 + \Delta E_3 = -1310 \text{ J}$$

$$W = W_1 + W_2 + W_3 = 449 \text{ J}$$

$$Q = Q_1 + Q_2 + Q_3 = 761 \text{ J}$$

$$\Delta E = Q - W = 312 \text{ J}$$



四、绝热过程——系统与外界无热量交换的过程

绝热过程 { 准静态绝热过程
非准静态绝热过程

1. 准静态绝热过程

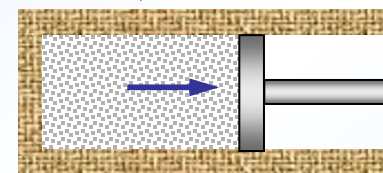
特征 $dQ = 0$ $dE + dW = 0$ $dW = -dE$

内能增量 $\Delta E = \nu \frac{i}{2} R \Delta T = \nu C_{V,m} \Delta T$

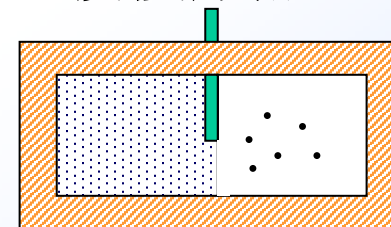
对外做功 $W = -\Delta E = -\nu C_{V,m} \Delta T$
$$= \frac{p_2 V_2 - p_1 V_1}{1 - \gamma}$$

吸热 $Q = 0$

结论: 当气体绝热膨胀对外做功时, 气体内能减少.

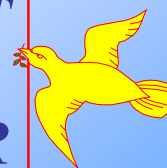


缓慢膨胀



自由膨胀

$$\begin{aligned} p_2 V_2 - p_1 V_1 &= \nu R \Delta T \\ (\gamma - 1) C_{V,m} &= R \end{aligned}$$



2. 理想气体准静态绝热过程的过程方程

$$dE = \nu \frac{i}{2} R dT = \nu C_{V,m} dT \quad dW = p dV$$

$$\because dE + dW = 0 \quad \therefore \nu C_{V,m} dT + p dV = 0 \quad (1)$$

在过程中任一时刻理想气体的状态满足

$$pV = \nu R T$$

$$\text{则有 } p dV + V dp = \nu R dT \quad (2)$$

从(1)、(2)中消去 dT ，得

$$(C_{V,m} + R) p dV + C_{V,m} V dp = 0$$

$$\text{即 } \frac{dp}{p} + \gamma \frac{dV}{V} = 0$$

积分可得 $\ln p + \gamma \ln V = \text{常量}$ 或 $pV^\gamma = C_1$ 泊松₁₃方程

$$C_{V,m} + R = C_{p,m}$$

$$\frac{C_{p,m}}{C_{V,m}} = \gamma$$



理想气体准静态绝热过程的过程方程

经推导可得

$$\left. \begin{aligned} pV^\gamma &= C_1 \\ TV^{\gamma-1} &= C_2 \\ p^{\gamma-1}T^{-\gamma} &= C_3 \end{aligned} \right\} \text{绝热过程方程}$$

或

$$\left\{ \begin{aligned} p_1 V_1^\gamma &= p_2 V_2^\gamma \\ T_1 V_1^{\gamma-1} &= T_2 V_2^{\gamma-1} \\ p_1^{\gamma-1} T_1^{-\gamma} &= p_2^{\gamma-1} T_2^{-\gamma} \end{aligned} \right.$$

$$W = \int_{V_1}^{V_2} p dV = \int_{V_1}^{V_2} C_1 \frac{dV}{V^\gamma} = \int_{V_1}^{V_2} p_1 V_1^\gamma \frac{dV}{V^\gamma} = p_1 V_1^\gamma \left(\frac{V_2^{1-\gamma} - V_1^{1-\gamma}}{1-\gamma} \right) = \frac{p_2 V_2 - p_1 V_1}{1-\gamma}$$

3. 等温线与绝热线的比较

等温过程方程

$$pV = \text{常量}$$

等温线的斜率

$$K_T = \left(\frac{dp}{dV} \right)_T$$
$$= -\frac{p}{V}$$

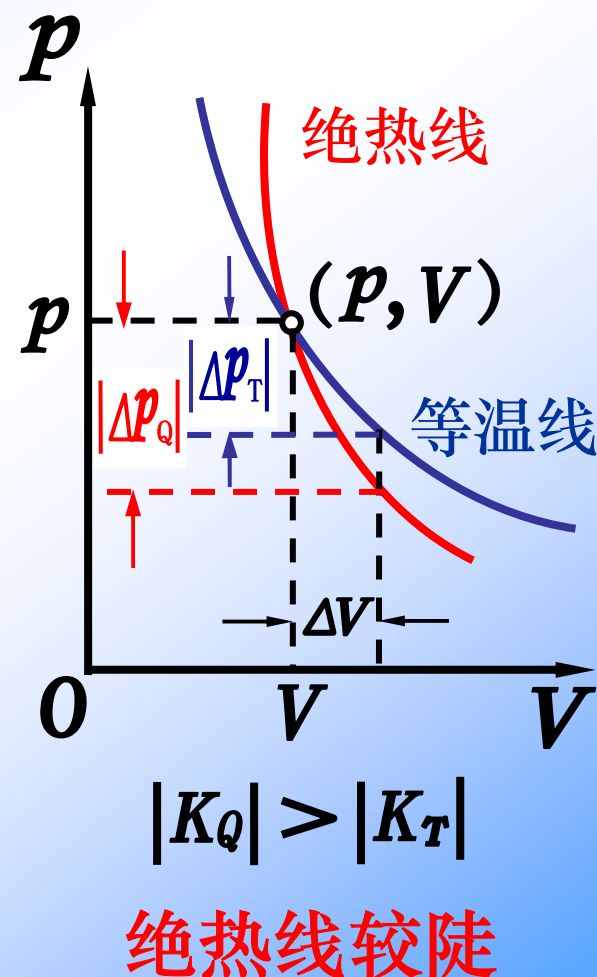
绝热过程方程

$$pV^\gamma = \text{常量}$$

绝热线的斜率

$$K_Q = \left(\frac{dp}{dV} \right)_Q$$
$$= -\gamma \frac{p}{V}$$

其中 $\gamma = \frac{c_p}{c_v} > 1$



绝热线比等温线陡峭的物理解释

考虑从 V_1 膨胀到 V_2 的准静态过程

等温过程：温度 T 不变

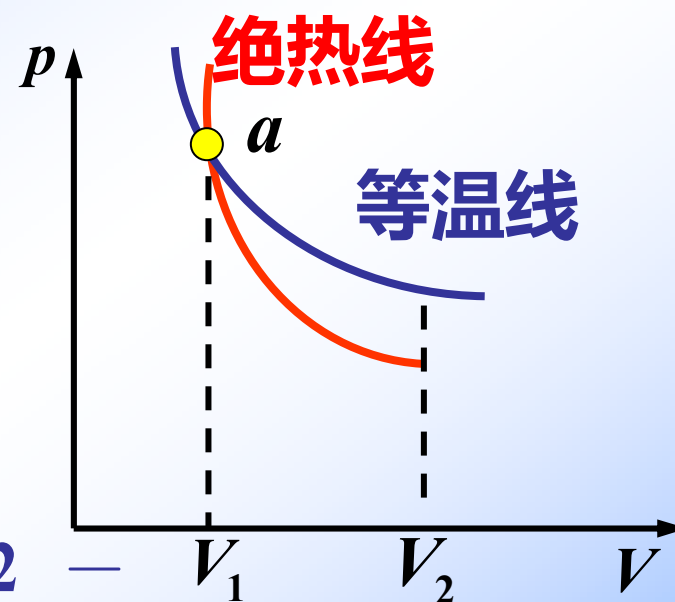
绝热过程： $Q = \Delta E + A = 0$

$$W = -\Delta E = -\nu C_{V,m} \Delta T$$

所以温度降低

等温： $\downarrow p \quad \uparrow V = \text{恒量}$
分子密度 $n \downarrow$ 而： $p = \frac{2}{3} n \overline{\varepsilon_t}$

绝热： $\downarrow \downarrow p \quad \uparrow V = \text{恒量} \quad \downarrow \overline{\varepsilon_t} = \frac{3}{2} kT$



可见，从相同初态 a 作同样的体积膨胀时，绝热过程的压强比等温过程的压强减少得多些。

过程	特征	过程方程	Q	W	ΔE
等容	$\Delta V = 0$	$\frac{p}{T} = C$	$\nu C_{V,m} \Delta T$	0	$\nu C_{V,m} \Delta T$
等压	$\Delta p = 0$	$\frac{V}{T} = C$	$\nu C_{p,m} \Delta T$	$p \Delta V = \nu R \Delta T$	$\nu C_{V,m} \Delta T$
等温	$\Delta T = 0$	$pV = C$	$Q = A$	$\nu R T \ln \frac{V_2}{V_1}$	0
绝热	$Q = 0$	$pV^\gamma = C$ $V^{\gamma-1} T = C$ $p^{\gamma-1} T^{-\gamma} = C$	0	$-\nu C_{V,m} \Delta T$	$\nu C_{V,m} \Delta T$

$$\gamma = \frac{C_{p,m}}{C_{V,m}} = \frac{(i+2)R/2}{iR/2} = \frac{i+2}{i} > 1$$

$$Q = W + \Delta E$$

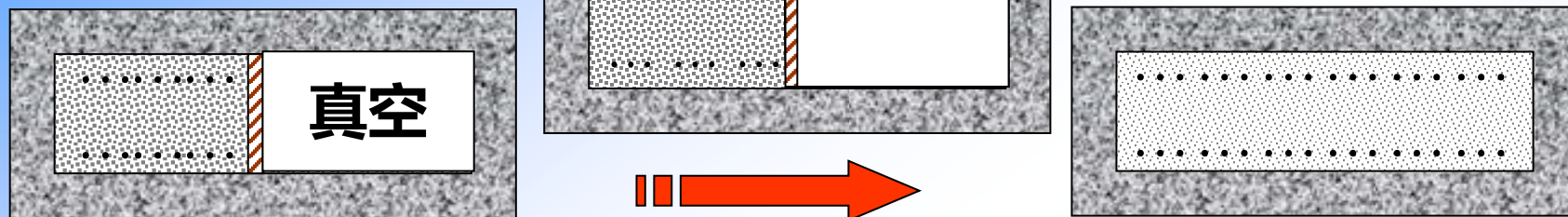
过程	特征	过程方程	Q	W	ΔE
等容	$\Delta V = 0$	$\frac{p}{T} = C$	$\nu C_{V,m} \Delta T$	0	$\nu C_{V,m} \Delta T$
等压	$\Delta p = 0$	$\frac{V}{T} = C$	$\nu C_{p,m} \Delta T$	$p \Delta V = \nu R \Delta T$	$\nu C_{V,m} \Delta T$
等温	$\Delta T = 0$	$pV = C$	$Q = A$	$\nu R T \ln \frac{p_1}{p_2}$	0
绝热	$Q = 0$	$pV^\gamma = C$ $V^{\gamma-1} T = C$ $p^{\gamma-1} T^{-\gamma} = C$	0	$\frac{p_2 V_2 - p_1 V_1}{1 - \gamma}$	$\nu C_{V,m} \Delta T$

$$\gamma = \frac{C_{p,m}}{C_{V,m}} = \frac{(i+2)R/2}{iR/2} = \frac{i+2}{i} > 1$$

$$Q = W + \Delta E$$

4. 非准静态绝热过程

绝热自由膨胀



1 (V_1, p_1, T_1)

2 (V_2, p_2, T_2)

自由膨胀过程中每个时刻都不是平衡态，但过程中：

$$W = 0, Q = 0, \quad \therefore \Delta E = 0, \text{ 则 } \Delta T = 0, T_2 = T_1$$

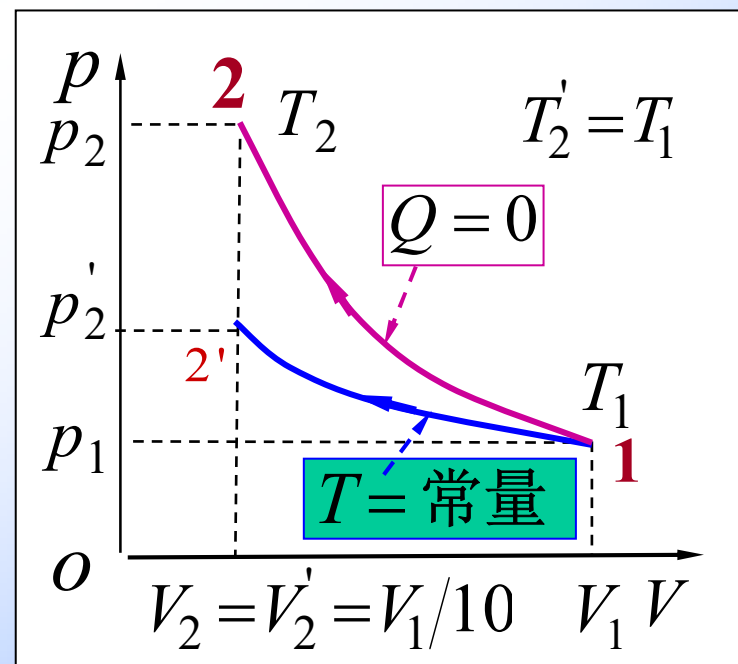
$$\left. \begin{array}{l} p_1 V_1 = \nu R T_1 \\ p_2 V_2 = \nu R T_1 \end{array} \right\} p_1 V_1 = p_2 V_2 \xrightarrow{V_2 = 2V_1} p_2 = \frac{1}{2} p_1 \quad ? \quad \cancel{p_2 = \frac{1}{2'} p_1}$$

注意

(1) 尽管 $T_2 = T_1$ ，但此过程不是等温过程。

(2) 由于是非准静态过程，所以绝热过程方程不适用。

例1 设有 5 mol 的氢气，最初温度 20°C ，压强 $1.013 \times 10^5 \text{ Pa}$ ，求下列过程中把氢气压缩为原体积的 $1/10$ 需作的功：（1）等温过程
（2）绝热过程 （3）经这两过程后，气体的压强各为多少？



已知: $\nu = 5 \text{ mol}$ $T_0 = 293 \text{ K}$

$$P_0 = 1.013 \times 10^5 \text{ Pa} \quad V = 0.1 V_0$$

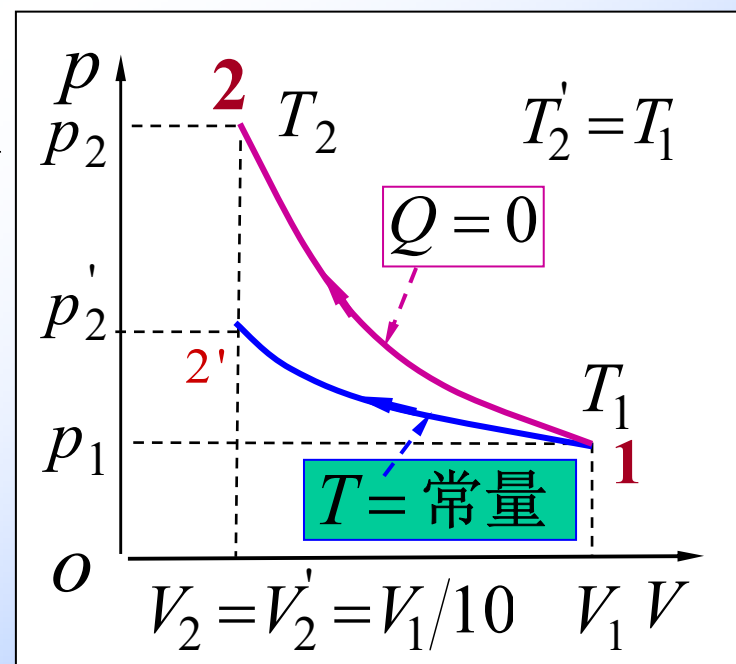
解 (1) 等温过程

$$W'_{12} = \nu RT \ln \frac{V'_2}{V_1} = -2.80 \times 10^4 \text{ J}$$

(2) 对刚性双原子分子

$\gamma = 1.4$, 有

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 736 \text{ K}$$



$$W_{12} = -\nu C_{V,m}(T_2 - T_1) \quad C_{V,m} = \frac{5}{2}R = 20.79 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$$

$$W_{12} = -4.60 \times 10^4 \text{ J}$$

(3) 对等温过程

$$p'_2 = p_1 \left(\frac{V_1}{V_2} \right)$$

$$= 1.01 \times 10^6 \text{ Pa}$$

对绝热过程, 有

$$p_2 = p_1 \left(\frac{V_1}{V_2} \right)^\gamma$$

$$= 2.54 \times 10^6 \text{ Pa}$$

